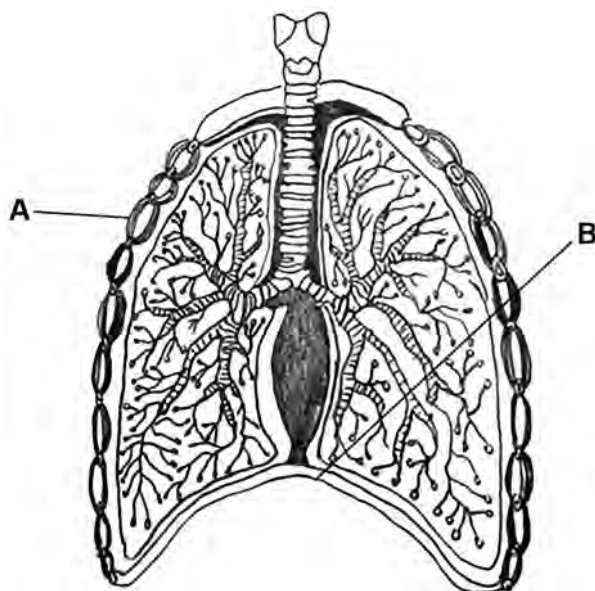


0 6

A student completed a dissection of an animal gas exchange system.

Figure 5 shows the scientific drawing he produced from his observations.

Figure 5



0 6

1

Suggest **two** ways the scientific drawing in **Figure 5** could be improved.

Do **not** refer to adding labels or using a sharp pencil in your answer.

[2 marks]

1 _____

2 _____

0 6 . 2

Describe the role of structures **A** and **B** in increasing the volume in the thoracic cavity.
[2 marks]

0 6 . 3

Lung volume can be determined with a spirometer using the helium dilution method.

The lung volume (V_2) is calculated using the equation:

$$V_2 = \frac{V_1(C_1 - C_2)}{C_2}$$

where

C_1 = the initial concentration of helium in the spirometer

C_2 = the helium concentration in the spirometer when equal with the lungs

V_1 = volume of gas in the spirometer.

A student used a spirometer with a volume of 3250 cm^3 and an initial concentration of 10% helium. They found C_2 was 3.78%

Use the equation to calculate this student's lung volume in dm^3

Show your working.

[2 marks]

Lung volume = _____ dm^3

Question 6 continues on the next page

Asbestos is a fibrous mineral. If asbestos is inhaled, it can cause a lung disease called asbestosis.

The latency period of a disease is the time between exposure to the cause of the disease and the symptoms occurring. The latency period for asbestosis is 15 to 40 years.

Asbestosis can be diagnosed using a biopsy. This is where a sample of lung cells is removed from a patient and examined with a microscope.

Asbestosis can also be diagnosed without a biopsy. To diagnose asbestosis without a biopsy, five conditions must be met. Three of these are given below.

1. A scan of the lungs showing fibrous scar tissue.
2. Reduced gas exchange.
3. Shortness of breath.

These three conditions are also present in many other lung diseases.

0 6 . 4

Using the information provided, suggest the other **two** conditions that must be met before a diagnosis of asbestosis can be made.

[2 marks]

1 _____

2 _____

0 6 . 5

Suggest **two** advantages of diagnosing asbestosis **without** a biopsy.

[2 marks]

1 _____

2 _____

Question 6 continues on the next page

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ANSWER IN THE SPACES PROVIDED**

Turn over ►

Idiopathic pulmonary fibrosis (IPF) is a lung disease similar to asbestosis. Both diseases cause:

- the death of alveolar epithelial cells due to DNA damage
- dead cells to be replaced by inelastic, fibrous scar tissue.

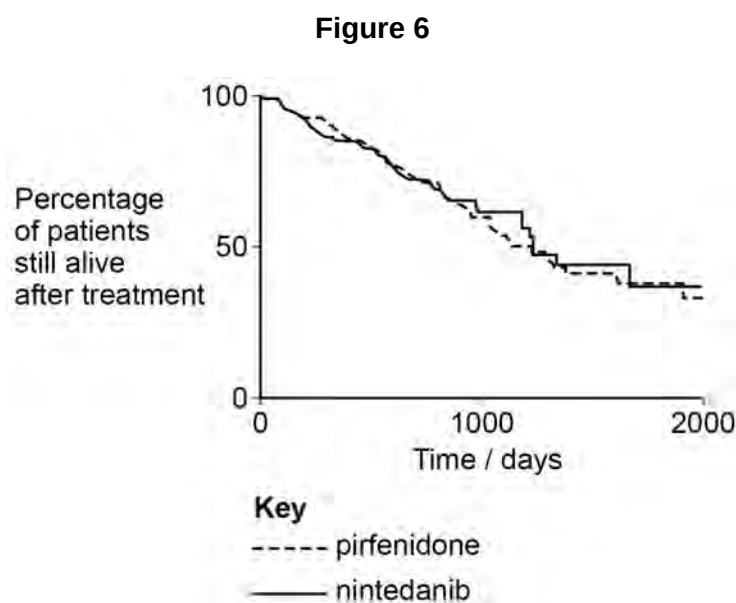
Currently, there is no cure for asbestosis and no effective treatment to slow its progression. There is also no cure for IPF. However, the drugs nintedanib and pirfenidone can slow the formation of scar tissue and slow the progression of IPF.

In two **separate** clinical trials, doctors investigated the effect of nintedanib and pirfenidone on 263 IPF patients.

- 124 patients were treated with nintedanib. 7 of these patients had to stop treatment due to side effects.
- 139 patients were treated with pirfenidone. 14 of these patients had to stop treatment due to side effects.

The doctors recorded how many days each patient lived after treatment.

Figure 6 shows their results.



0 6 . 6 Suggest why neither clinical trial had a control group.

[1 mark]

Section B

Answer **one** question.

You are advised to spend no more than 45 minutes on this section.